

Advanced SQL: Null Functions

Question 1

Select

employee-id,
name,
IFNULL(salary, 0) AS salary-with-default
FROM Employees;

Employees

employee-id	name	salary-with-default
1	Alice	5000
2	Bob Charlie	0
3	Charlie	7000
4	Dana	0

Question 2

Select

employee-id,
name,
IFNULL(department, "Not Assigned") AS ~~department~~ AS department-name
FROM Employees;

Employees

employee-id	name	department-name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Question 3.

Select

~~custo~~ order-id

customer-id

FROM Orders

WHERE ISNULL(customer-id);

Orders

order-id	customer-id
101 103	NULL
102	
103	

Question 4.

SELECT

order-id,

customer-id

IFNULL(delivery-date, 'Pending')

AS delivery-status.

FROM Orders;

or

Orders

order-id	customer-id	delivery-status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Question 5

SELECT

student-id,
name

IFNULL(grade, 0) AS final-grade
FROM Students;

*Students

student-id	name	final-grade
1	Ethan	85
2	May	0
3	Olivia	90

Question 6

SELECT

grade Count(student-id) AS not-graded-count
FROM Students
~~HAVING grade = 'NULL';~~
WHERE grade IS NULL;

not-graded-count
1

Question 7

SELECT

product-id,
name,

price - IFNULL(discount, 0) AS final-price
FROM Products;

Products

product_id	name	final-price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

Question 8.

SELECT

COUNT(*) AS missing_email_count
FROM Customers
WHERE email IS NULL;

missing_email_count
2

Question 9.

SELECT

customer_id,
name,
IFNULL(email, "No Email") AS email_display
FROM Customers;

Customers

customer_id	name	email
1	Linda	No Email
2	Joseph	joseph@mail.com
3	N/A	No Email

Question 10

Select

payment_id,
IFNULL (method, 'Unknown') AS method_display,
status

FROM Payments;

Payments

payment-id	method_display	status
301	sp Credit	NULL
302	Paypal	Success
303	Unknown	Failed

Question 11

Select item-id,

item-name,

IFNULL (quantity, 0) AS quantity_checked

FROM Inventory;

Inventory

item-id	item-name	quantity
1	Pen	0
2	Notebook	150
3	Eraser	0

Question 12.

Select emp-id,
COALESCE (bonus, commission) AS first-available-reward
FROM Employees-Extra;
Employees-Extra

emp-id	first-available-reward
1	300
2	100
3	NULL

Question 13

Select COUNT(*) AS no-room-count
FROM Classes
WHERE room IS NULL;

no-room-count
2

Question 14 :

SELECT student-id,
date

IFNULL(status, 'Not Marked') AS attendance-status

FROM Attendance;

Attendance

student-id	date	status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Question 18

SELECT

account-id
IFNULL(account-type, 'Unknown') AS type-display,
IFNULL(balance, 0) AS balance-checked
FROM Bank-Accounts;

~~account-~~

Bank-Accounts

account-id	account-type	balance-checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Question 16

SELECT project-id,
title,

IFNULL(CAST(start-date AS STRING), 'TBD')
AS start-display

FROM Projects;

project-id	title	start-date
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Question 17

Question 17

```
SELECT review-id,  
       product-id,  
       IFNULL(comment, 'No Comment') AS 'comment display',  
       IFNULL(rating, 0) AS rating-display  
FROM Reviews
```

review-id	product-id	comment	rating
1	501	Great product	4
2	502	No Comment	0
3	503	Works fine	3

Question 18,

```
SELECT supplier-id,  
       name,  
       COALESCE(phone, alt.phone, 'No Contact')  
       AS contact-number.
```

review-id	product-id	contact-number
1	501	123456789
2	502	987654321
3	503	No Contact.

Question 19

```
SELECT user_id,  
       IFNULL(theme, 'light') AS theme_set,  
       IFNULL(language, 'English') AS language_set,  
       IFNULL(timezone, 'UTC') AS timezone_set  
FROM User_Settings;
```

User_Settings.

user_id	theme	language	timezone
1	light	English	UTC
2	Dark	English	UTC+1
3	light	English	UTC

Question 20

```
SELECT record_id,  
       machine_id,  
       IFNULL(issue, 'Unknown Issue') AS issue_log,  
       IFNULL(technician, 'Not Assigned') AS technician_name  
FROM Maintenance;
```

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed	Alex